# Aircraft Variants for the 2027 Fly-off

Two complete aircraft, built from every dataset the team holds, and the reasoning behind each choice. Written for the team to test, challenge and sign.

SITE TIMPA, TUCSON AZ
32.2647 N 111.2739 W
ELEV 667 M
DESIGN AIR 1.064 KG/M$^3$
FLY-OFF 15-18 APR 2027
CONFIG FREEZE 19 FEB 2027

VARIANTS IN THIS DOSSIER

A

## Fowler-flap compact carrier

0.9 m² wing with Fowler flaps, single 8S tractor, one dense sensor declared at tech. Recommended.

P(TOP OF FIELD) 0.286
MEAN TOTAL 6.76
DB-01 CONTROL 0.128

B

## Slotted-flap broad-wing carrier

1.0 m² wing with simple slotted flaps; wins where the Fowler flap disappoints; shares every long-lead part.

P(TOP OF FIELD) 0.238
MEAN TOTAL 6.62
SWITCH GATE 15 JAN 2027

| DOCUMENT | REVISION | PREPARED BY | STATUS |
|---|---|---|---|
| AUM-DBF-27-DB-002 | A, 30 October 2026 | Badr Obtel | Recommendation; team decision pending |

01

# Decision register

### How to read it

1.1

Every decision in this dossier appears once here and once as a card in the body, under the same identifier. The register gives the recommendation, the strength of its evidence, what would change it and when it freezes. The shaded column is left for the team to sign; nothing here is decided until it is signed.

Table 01

The decisions this dossier recommends. Grades: `VERIFIED` verified source, `COMPUTED` our validated model, `ESTIMATE` reasoned estimate.

| ID | Recommendation | Grade | Changes if | Freezes | Team |
|---|---|---|---|---|---|
| P-01 | Declare the sensor mass at technical inspection, planned for a hot day | COMPUTED | Q&A fixes it earlier | Tech, Apr 2027 | |
| S-01 | Waive Mission 1 | COMPUTED | Q&A denies the credit | Fly-off | |
| S-02 | One tungsten dart rated to 150 g, tow eye above its CG | ESTIMATE | TC-02 failures below 150 g | CDR | |
| S-03 | One container (re-cut pads) and adjustable simulators, 3 to 6 kg | ESTIMATE | TC-02 | CDR | |
| S-04 | Closed-loop drum with slip ring; ballast chosen so the stow fits | COMPUTED | Stow ruled outside the window | CDR | |
| S-05 | Drop from 1.524 m | VERIFIED | TC-02 failures at 1.524 m | GM | |
| S-06 | SD7037 airfoil | VERIFIED | Better tunnel-tested section | 19 Feb | |
| S-07 | Three LED arrays, one-button sequencing | ESTIMATE | TC-03 below 90 % | CDR | |
| C-01 | High-lift flap (Fowler on A, slotted on B), not a plain flap | COMPUTED | Flap gain below 0.1 | 19 Feb | |
| C-02 | Keep DB-01's drag; hold the lower estimate in reserve | ESTIMATE | First measured flight drag | First flight | |
| E-01 | No RQ-4 proportions; styling only where free | COMPUTED | – | – | |
| A-01 | Variant A: 0.9 m² wing with Fowler flaps | COMPUTED | TC-07 below 1.75 | 15 Jan | |
| A-02 | Bay and tow exit on the CG; 16 % wing-tail margin | COMPUTED | Weigh-in CG off by 3 % | CDR | |
| A-03 | Single 8S AT7215 / 20x15E; 80 A blade fuse rated 58 V | VERIFIED | TC-06 fuse duty | 19 Feb | |
| A-04 | High wing, conventional tail, fixed faired tricycle | COMPUTED | Pilot prefers a taildragger | 19 Feb | |
| B-01 | Variant B: 1.0 m² wing with slotted flaps, if A fails its gate | COMPUTED | TC-07 passes | 15 Jan | |
| B-02 | A and B share every long-lead part | – | – | Now | |
| H-01 | Build A; switch to B only on the dated gate | COMPUTED | TC-07 result | 15 Jan | |

02

# Summary

### The answer

2.1

Build **Variant A**: a 0.9 m² straight wing with Fowler flaps on the full 1.82 m span, a high wing over a box fuselage, a conventional tail, fixed tricycle gear and a single 8S tractor, carrying one dense sensor whose maximum mass is declared at technical inspection between 3 and 6 kg. Keep **Variant B**, the same aircraft with a 1.0 m² wing and simple slotted flaps, one wing away, and choose between them on 15 January 2027 on the result of a Fowler-flap bench test.

| P(top), A | P(top), DB-01 | Mean total, A | Stall at M2, A | Declared sensor | P(1st), report 90 |
|---|---|---|---|---|---|
| 0.286 | 0.128 | 6.76 / 7.5 | 12.2 m/s | 3–6 kg | 0.348 |

### Why, in five sentences

2.2

1. Most of the 2027 score is paid for finishing, and teams get only two or three flight attempts, so every choice is judged first by what it does to the chance of completing Missions 2 and 3.
2. The declared sensor mass feeds the Ground Mission and Mission 2 together; choosing it at technical inspection, once the real aircraft has been weighed and flown, is worth about four tenths of a point and is the single largest gain in this study.
3. A heavy sensor is only an asset if the aircraft can land it slowly; at the 1.82 m span the way to do that without a big, slow wing is a high-lift flap, and the Fowler flap gives Variant A a Mission 2 stall speed of 12.2 m/s against 13.5 m/s for the DB-01 design.
4. Placing the payload bay and the tow-line exit on the centre of gravity makes every loading fly the same, which matters more than any aerodynamic refinement for a hand-flown aircraft.
5. The aircraft makes winning possible, but the design report decides it: Variant A's chance of finishing first rises from 0.116 at a report score of 80 to 0.348 at 90.

### What changed from DB-01, and what did not

2.3

DB-01's direction survives: a straight flapped wing near 0.9 m², a single tractor, a high wing, a conventional tail, fixed gear, a closed-loop drum and a waived Mission 1. Four things changed. Its plain flap reached less lift than assumed, and a high-lift flap replaces it. Its container broke the drop-height envelope, and a tougher, denser sensor fixes that. Its frozen 4 kg payload unit gives way to a declaration made at technical inspection. And its 100 A fuse was never rated for an 8S pack, so a 58 V, 80 A part replaces it.

### What the team must sign

2.4

The decision register (Table 01) lists eighteen recommendations. Three need action now: order the long-lead parts common to both variants, build the Fowler test section (TC-07), and file the five round-one rules questions by their deadline (Section 13).

# 01

# The problem

What the 2027 rules reward, what changed since DB-01, and how every design in this dossier was scored.

IN THIS CHAPTER

01 What wins in 2027
02 What changed since DB-01
03 Evidence and method

03

# What wins in 2027

### The score in one line

3.1

Every flight and ground mission is scored against the best result any team posts that year, our own included:

$$\text{Total Mission} = M_1 + \underbrace{1 + \frac{W/t}{\max(W/t)}}_{M_2} + \underbrace{2 + \frac{N_{\text{laps}}\, W_s}{\max(N_{\text{laps}}\, W_s)}}_{M_3} + \underbrace{0.5 + \frac{W_s\, h}{\max(W_s h)}}_{GM} \;\leq\; 7.5$$

Three facts follow directly from the rule text, and they shape everything in this dossier.

- **Most of the score is paid for finishing.** Completing the four missions earns $1 + 1 + 2 + 0.5 = 4.5$ of the 7.5 points before any performance bonus. The bonuses add at most 3.0, and each is shared out as a ratio to the field's best.
- **One number appears three times.** The declared sensor mass $W_s$ sets the Ground Mission product, sets the mass of every Mission 2 payload unit (container plus maximum sensor, rules §3.1.3), and caps the Mission 3 ballast. The rules let the team choose that number at technical inspection in April (§4.1), after the aircraft has flown.
- **The best result is a moving target that includes ours.** If we set the maximum in a mission, our ratio is exactly 1 and every rival's shrinks.

### Finishing beats speed

3.2

The fly-off results since 2022 make completion the dominant term. In 2026, 83 teams passed technical inspection and only 17 completed every mission. Between 2022 and 2026 the share of Mission 2 finishers who went on to finish Mission 3 ranged from 0.35 to 0.66, and teams that passed inspection received only 2.4 to 2.9 official flight attempts each.[1] DB-01's trade-weight study gave mission completion 55 % of the weight across all design criteria, more than three times the 17 % given to turn and lap performance.[2]

Consequence

**Every design choice in this dossier is judged first by what it does to the chance of finishing Missions 2 and 3, and only then by the bonus it buys.**

### The report multiplies everything

3.3

The competition score is the Total Report Score multiplied by the Total Mission Score. At a report score near 88 and a mission score near 6.3, one mission point is worth 14.0 report points. Between 2022 and 2026 a team with a report score of 88 needed a mission total of about 5.66 to win.[3] The aircraft decides the mission score; the design report decides the multiplier. Neither can carry the other.

[1] `analysis/data/field_stats.csv` and `attempts.py`, built from the AIAA competition summaries 2022–26; verified against the source tables on 13 October 2026.

[2] `analysis/out/s6_trade_weights.csv`: each criterion weighted by the change in total score for a ±10 % change in it.

[3] Stream S1, `out/rules/results.json`, computed from `analysis/data/dbf_results.csv` (513 rows).

### Where the bonus points are

3.4

Each bonus has a different shape, and the shapes point to one strategy.

- **Ground Mission.** Linear in the declared sensor mass at a 1.524 m drop. A 6 kg sensor gives 9.14 kg m, the expected-field anchor; 8 kg gives 12.2 kg m, the aggressive anchor. Above that the bonus is saturated.
- **Mission 3.** Laps times towed mass, inside five minutes. Laps come in whole numbers, so the product jumps: seven laps at 4.4 kg gives 30.8 kg laps against an aggressive-field anchor of 27.
- **Mission 2.** Payload per second over five laps. The DB-01 baseline carried 8 kg in 207 s, 0.0386 kg/s, against an expected-field anchor of 0.067 kg/s. This is the hardest bonus to lead, because it needs a heavy aircraft that is also fast.

Reading

**A heavy declared sensor feeds two bonuses at once and costs Mission 2 almost nothing, provided the aircraft can still land it. That proviso is the design problem.**

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P-01 / WORKED EXAMPLE / EXPECTED FIELD / 120 PAIRED DRAWS

## P-01

## A worked example: the same aircraft, two ways of declaring the sensor

Recommendation

**Choose the declared sensor mass at technical inspection, not now.**

Why, plainly

DB-01 froze a 4 kg payload unit at design time, so a heavy sensor looked dangerous: if the finished aircraft came out heavy, the unit could not be lightened and Mission 2 would fail. Declaring in April, after the aircraft has been weighed and flown, removes that gamble. The team declares as heavy as the real aircraft can safely carry on a hot afternoon.

The engineering

On the same 0.9 m² aircraft and the same uncertain inputs, freezing the unit with the same hardware scores a mean of 6.30 to 6.34, while choosing the declaration at tech scores 6.74 for a 3 to 6 kg range and 6.76 for 3 to 8 kg. The gain comes almost entirely from the Ground Mission, which rises from about 1.0 to 1.4, while Mission 2 and Mission 3 hold. Planning the declaration for an average day instead of a hot one lets heavy declarations fail on hot afternoons and drops the worst-tenth score to 2.5.

Derivation

Per draw $i$ of the uncertain inputs: $m_i^{*} = \arg\max_{m \in [3,\, m_{hi}]} \text{Total}(m \mid \text{inputs known at tech},\ \rho = 1.04)$, then score $\text{Total}(m_i^{*} \mid \text{draw } i)$. Inputs known at tech: airframe mass, $C_{L,\max}$, propulsive efficiency, container mass, our measured lap factor. Not known: the day's density and wind, the field.

What would change it

A Q&A ruling that fixes the declared mass before April, or a container that cannot protect the sensor across the whole range (TC-02).

04

# What changed since DB-01

### Why this chapter exists

4.1

DB-01 (14 October) recommended a 0.9 m² straight flapped wing, a 4 kg payload unit, a waived Mission 1, a high wing, a conventional tail, fixed tricycle gear, a single tractor and a closed-loop drum, with a mean of 6.33 of the 7.5 available mission points. None of it was signed. Since then five evidence streams have tested its inputs. Some held, some were optimistic, and one change of method moved the answer more than all the corrections together. Table 02 lists them; the cards that follow explain the three that matter most.

Table 02

Every DB-01 input that new evidence moved. Effects are on the DB-01 design's mean mission score over the same 120 paired draws unless stated.

| Input | DB-01 assumed | New evidence | Effect |
|---|---|---|---|
| Landing $C_{L,\max}$, plain flap | 1.60 | 1.44 trimmed, area method validated on NACA TN 663 COMPUTED | about −0.15; recovered by slotted or Fowler flaps |
| Clean $C_{L,\max}$ | 1.25 | 1.19 COMPUTED | limits turn lift |
| Oswald $e$ | 0.946 (Raymer) | 0.901; same method gives 0.868 where ERAU measured 0.842 COMPUTED | −0.05 |
| Parasite drag $C_{D0}$ | 0.0230 | build-up 0.0161 ESTIMATE, not banked (card C-02) | 0 (upside held back) |
| Container | 0.6 kg, 100 g sensor | 100 g container breaks the 381 mm drop envelope (0.449 m); 150 g sensor, 0.63 to 1.18 kg ESTIMATE | sensor must be built to 150 g |
| Payload unit | frozen at 4 kg | declared at tech, 3 to 8 kg, one container COMPUTED | +0.40 to +0.43 |
| Reliability per attempt | flat 0.65 | depends on stall speed, thrust and complexity ESTIMATE | reorders designs |
| Field | two fixed scenarios | 5,000 simulated fields, calibrated on 2022–26 ESTIMATE | enables P(1st) |

C-01 / EVIDENCE COMPUTED / UIUC, NACA TN 663, TR 534, TR 664

C-01

## Maximum lift was optimistic, and the fix is a better flap

### Recommendation

**Use a high-lift flap (slotted or Fowler), not a plain flap.**

### Why, plainly

The slowest safe landing speed sets how heavy the aircraft can be in Mission 2. DB-01 assumed the wing could reach a lift coefficient of 1.6 with a simple hinged flap. Wind-tunnel data for our airfoil, run through a method that we checked against twelve measured wings, say 1.44 once the tail's balancing load is included. That gap costs about 0.15 points. A flap that slides back and opens a slot recovers it and more.

### The engineering

Trimmed landing $C_{L,\max}$ at 0.9 m²: plain 30 % flap 1.44, single-slotted 1.62, Fowler 1.92. The section increments come from Williamson's UIUC flap tests (plain, mea-

sured at $Re = 3 \times 10^5$), Wenzinger and Harris NACA TR 664 (slotted) and Platt NACA TR 534 (Fowler). The wing value uses the area-ratio method, which never over-predicted on NACA TN 663's twelve measured wings.

Derivation

$\Delta C_{L,\max,\text{wing}} = 0.9\,K\,\Delta C_{l,\max}\,S_{\text{flapped}}/S$, with $K = 1.07$ (TN 663 calibration). Trim: $C_{L,\text{trim}} = C_{L,w}\,(1 - \bar{c}\,C_{m,ac}/(l_t\,C_{L,w}))$ with $l_t = 3\bar{c}$.

What would change it

A wing-model test or flight test showing the slotted or Fowler gain below 0.1 in $C_{L,\max}$.

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C-02 / ORCHESTRATOR JUDGEMENT / UPSIDE HELD IN RESERVE

## C-02

### A lower drag estimate exists, and we deliberately do not use it

Recommendation

**Keep DB-01's drag ($C_{D0} = 0.0086 + 0.013/S$) as the central value.**

Why, plainly

A careful component-by-component estimate says the aircraft could have about 30 % less drag than DB-01 assumed. Banking that would raise every score. We do not, for two reasons. When the same method was applied to the RQ-4-style model we also ran through CFD, it under-predicted the body and nacelle drag by about a third. And the lap-time factor the model already applies, taken from 24 flights where teams compared their predictions with reality, exists precisely because drag estimates like ours come out optimistic.

The engineering

Build-up $C_{D0}$ 0.0161 (range 0.0135–0.0209) against DB-01's 0.0230; non-wing drag area 0.009 m² against 0.013. RQ-4 cross-check: the build-up gives 0.0042 m² of non-wing drag area where the CFD gives 0.0067 m². If the build-up proved right in flight, the 0.9 m² design would gain about 0.09 points and the 1.0 m² design about 0.46.

What would change it

Measured drag from the first flight tests (power at known speed and mass), which closes this question within a week of first flight.

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# 05 Evidence and method

### What the numbers rest on

5.1

Every number in this dossier traces to a result file in the DB-02 work folder, and each result carries its source and one of four grades.

The evidence base is the one DB-01 built, extended:

- **Contest record.** The 2026–27 rules (draft of 2 September 2026, the latest official version), 99 of whose clauses are quoted verbatim in a compliance matrix and machine-checked against the text; 1,008 verified values from the AIAA top-3 design reports of 2021–26; 513 rows of fly-off results from 2022–26.

Table 03

Evidence grades used throughout. The tag sits beside the number it qualifies.

| Tag | Meaning |
|---|---|
| VERIFIED | read in a primary source (rules text, NACA or NASA report, UIUC tunnel data, manufacturer datasheet) and checked against it |
| RESELLER | taken from a retail listing only |
| COMPUTED | produced by our own model, which was itself checked against measured data (named where used) |
| ESTIMATE | a reasoned range with its derivation; sampled across that range in the Monte Carlo |

- **Aerodynamics.** UIUC low-speed airfoil tests (SD7037 and flap data), NACA TN 663 (twelve measured wings, used to validate the lift method), NACA TR 534 and TR 664 (Fowler and slotted flaps), XFoil 6.99 and AVL 3.52, and the team's own OpenFOAM simulation of the RQ-4-style concept.
- **Propulsion and payload.** APC and T-Motor measured performance maps calibrated on ERAU's 2023 flight; MIL-HDBK-304B cushioning curves for the container.
- **Site.** 30 years of mid-April weather at Tucson and the TIMPA elevation, which set the design air density of 1.064 kg/m³ (a hot afternoon) and the wind.

## How a design is scored

5.2

A candidate is scored the way it will actually be flown, in four steps, over 120 paired draws of every uncertain input (the same draws for every candidate, so differences between candidates are not noise).

1. **Build.** The draw fixes what the finished aircraft turns out to be: airframe mass (fitted to fourteen top-3 aircraft, ±31 % at one standard deviation), maximum lift, propulsive efficiency, drag, container mass, and how much slower than predicted the laps really are (24 measured flights, median ×1.15).
2. **Declare.** At technical inspection the team knows the built aircraft but not the weather of the day or the field. It declares the heaviest sensor that maximises the expected score on a hot afternoon ($\rho = 1.04$ kg/m³).
3. **Fly.** Missions are flown with that declaration under the draw's real conditions. Each attempt succeeds with a probability that falls as stall speed rises and thrust falls, and with every added mechanism.
4. **Compete.** Each of our draws is matched against simulated fields of about 80 rival teams, whose report scores, completions and results are drawn jointly and calibrated on the 2022–26 fly-offs. The mission maxima are recomputed with our result included.

What is ranked

**Designs are ranked by the probability that their mission total is the highest in the field. A selection rule was fixed on 27 October, before any result existed (Section 5.3).**

## The selection rule

5.3

- **Gates.** Every compliance row passes; the probability of finishing both Mission 2 and Mission 3 is at least 90 % of the best candidate's; the worst-tenth score is no lower than the DB-01 control's.
- **Variant A.** The gated candidate with the highest probability of the top mission total. If two candidates differ by less than the Monte Carlo uncertainty, the simpler one wins.
- **Variant B.** Physically different from A in a feature the rules freeze at the design report (gear type, tail type, wing configuration, tractor or pusher, number of motors), chosen as the best performer in the conditions where A does worst.

### What the model does not know

5.4

- The field's 2027 maxima are estimates anchored on 2021 and 2025 results. They are drawn over a wide range, and every comparison between our candidates uses the same draws.
- The per-attempt reliability rests on thin data: 2022–26 attempt statistics and a handful of documented failures. Its coefficients are graded estimates and are varied in the regret analysis.
- Within one draw the scoring code still chooses the number of simulators flown and the Mission 3 ballast with knowledge of that draw. This flatters every candidate about equally; the declaration itself is chosen without that knowledge.
- Pilot skill, build quality and the design report enter only through reliability and schedule. They are where a real team wins or loses.

# 02

# The design space

Every family the team proposed or the rules allow, and the number that removed each one.

IN THIS CHAPTER


| | |
|---|---|
| 04 | The RQ-4 reference family, in CFD |
| 04 | Families removed by the numbers |
| 04 | The integrated screen |

06

# The design space, and what was ruled out

**How wide the search was**

6.1

The search covered every aircraft family the team proposed or the rules allow, then every combination of wing area (0.8 to 1.3 m²), flap type (plain, slotted, Fowler), declared-sensor range (3 to 4.4, 6 or 8 kg) and propulsion (single 8S, single 12S, twin) on the families that survived. Each family below was removed for a stated reason, and each reason is a number.

**The RQ-4 reference family**

6.2

The team's starting reference was the MQ-9, MQ-4C and RQ-4 family of long-wing surveillance aircraft (Badr Ob, 10 October 2026), and the team modelled a scaled RQ-4-style concept in CAD and ran it through CFD. That work is the clearest single piece of evidence in this dossier for why the 2027 aircraft has to look different.

E-01 / EVIDENCE COMPUTED (CFD, INDICATIVE) / DECIDES WING SHAPE

E-01

## Keep the RQ-4 look where it is free; drop its proportions

Recommendation

**Do not use a long, narrow RQ-4-proportioned wing. Body styling may borrow from the family where it costs nothing.**

Why, plainly

A long, slender wing is the right answer for staying airborne for a day. This contest asks for the opposite: a heavy load lifted at a slow, safe landing speed and flown hard around tight turns, with the span capped at 1.83 m. At that span a long, slender wing has very little area. The CFD of the team's own RQ-4-style model shows the result: the whole aircraft cannot produce much more lift than a coefficient of 0.9, so carrying even 7 kg it would have to land at about 25 m/s, roughly twice the speed of a safe touchdown.

The engineering

At the 1.82 m span the RQ-4-style wing has $S = 0.232$ m² and an aspect ratio of 14.3. OpenFOAM gives a lift coefficient of 0.49 at 0° body angle, 0.786 at 4° and 0.896 at 12°, where the wing is fully stalled. With $C_{L,\max} \approx 0.89$, a 7 kg aircraft stalls at 24.9 m/s. The design study's wings carry 0.8 to 1.3 m² on the same span, 3.4 to 5.6 times the area.

Derivation

$$V_s = \sqrt{2mg/(\rho S C_{L,\max})} = \sqrt{2 \times 7 \times 9.807/(1.064 \times 0.232 \times 0.896)} \approx 24.9 \text{ m/s}.$$

What would change it

Nothing within these rules: the span cap and the landing-speed requirement fix the wing area, whatever the styling.

SURFACE PRESSURE / BODY 4° / 26.5 M/S



SPEED ON MEAN CHORD / BODY 12° / STALLED



**Figure 01**

The RQ-4-style concept in CFD at 26.5 m/s. Left: surface pressure at 4° body angle, where the aircraft reaches its best lift-to-drag ratio of 10.3. Right: speed on the wing's mean section at 12°, where the flow has left the upper surface from about 10 % chord; the wing is stalled at a lift coefficient of about 0.9.

DATA QUALITY / CFD, INDICATIVE

OpenFOAM 14, steady RANS with the $k$–$\omega$ SST $\gamma$–$Re_\theta$ transition model, half model, 2.58 million cells. Mean $y^+$ is 5.80 against a target of 1.5; the relaxed boundary-layer pass fails three of checkMesh's extra quality tests; the two stalled angles ended at the 1,000-iteration cap with a 1.2 to 1.9 % spread in forces; the hollow CAD shell leaked a small volume of flow into the fuselage. The sweep has four of nine angles complete. The results are used here for one purpose, the order of magnitude of the maximum lift, where they agree with the airfoil and wing data. They are not used to calibrate maximum lift or drag anywhere in the scoring.

## Families removed by the numbers

6.3

Table 04 lists every family removed from the search and the number that removed it. Each was scored on the same 120 paired draws as the finalists.

Table 04

Every family removed from the search, with the number that removed it. Score changes are on the mean mission score over the same 120 paired draws.

| Family | Why it was removed | Evidence |
|---|---|---|
| Delta, flying wing, blended body | Low maximum lift without a tail, high landing attitude: 4.60 against 6.29 for a straight wing of the same area | DB-01, COMPUTED |
| Biplane and box wing | Induced drag falls to 77 % of a monoplane's, but halved chords, struts and lower lift per area cost 0.32 points | S3 kill gate, COMPUTED |
| Tandem wing | No induced-drag gain at zero vertical gap (Munk's stagger theorem) | theory |
| Winglets and tip plates | +0.015 to +0.041 in *e* at heights of 5 to 10 % of the span; drag of the added surface and 0.10 to 0.15 kg cost more than they return: −0.01 to −0.05 | S3, AVL, COMPUTED |
| Flap in the turns | Raises turn lift from 1.18 to 1.29 but costs 5 % of span efficiency: −0.04 | S3, COMPUTED |
| Pusher propeller | The tow line and sensor trail behind and below the aircraft, through the pusher's disc during deployment and recovery | DB-01 risk R9 |
| Retractable gear | Pays only if lighter than 120 to 240 g; saves about 0.01 points against a gear-up landing risk | DB-01, COMPUTED |
| Frozen heavy payload unit | Superseded by declaring at technical inspection (card P-01) | S8, COMPUTED |

# 03

# Shared decisions

The payload, the strategy and the airfoil: everything both aircraft carry and do the same way.

IN THIS CHAPTER

05 Mission 1 waiver
05 Sensor, container and simulators
05 Deploy and recovery
05 Drop height, airfoil, lights

07

# Decisions both variants share

**Why these come first**

7.1

The two variants differ in how they are built, but they fly the same missions with the same payload hardware and the same strategy. Fixing these decisions once lets the variant chapters concentrate on the aircraft. Each card gives the recommendation, the reasoning in plain terms, the numbers, and what would change it.

S-01 / EVIDENCE COMPUTED / CONFIDENCE HIGH / DECIDED PER CONTEST, APRIL 2027

S-01

## Skip Mission 1 and take its credit through Mission 2

Recommendation

**Waive Mission 1 at the flight line.**

Why, plainly

Flight attempts are scarce: teams get two or three. Mission 1 is worth exactly one point, and a successful Mission 2 now awards that same point automatically. Flying Mission 1 spends an attempt to win something we are given anyway, and every attempt spent is one fewer chance to finish Mission 3, which is worth three.

The engineering

At the field's typical 65 % success per attempt and three attempts, waiving Mission 1 raises the probability of completing Mission 3 from 0.40 to 0.72 COMPUTED. Across the range of plausible success rates the waiver is worth up to 1.05 expected points. The cost is one fallback: if Mission 2 never succeeds, there is no Mission 1 score to keep.

What would change it

A Q&A ruling that the Mission 2 credit does not apply, or an aircraft whose Mission 2 success rate on test falls well below its Mission 1 rate.

S-02 / EVIDENCE ESTIMATE AND COMPUTED / MIL-HDBK-304B, BARROWMAN / FREEZE AT CDR

S-02

## Build one heavy, dense sensor rated to 150 g

Recommendation

**A tungsten-ballasted dart, about 200 mm long and 70 mm across, rated to survive 150 g, with its tow eye above its centre of gravity.**

Why, plainly

The sensor must survive six drops onto concrete from 1.524 m inside a container whose lifting ring may sit no more than 381 mm above its base. A softer sensor needs thicker foam, and thicker foam pushes the container past 381 mm. DB-01's container, sized for a 100 g sensor, already fails that limit. Making the sensor tougher and denser makes everything around it smaller: the container, the bay and the fuselage.

The engineering

The DB-01 container's worst ring-to-bottom distance is 0.449 m against the 0.381 m limit, and every 100 g design from 3 to 10 kg fails (385 mm at best). At 150 g, 41 mm of polyethylene foam suffices and the container passes up to 8 kg. A 6 kg tungsten sensor is 69.9 mm in diameter (lead would need 85.6 mm, steel 103 mm). Towed

from its nose it would hang 77.0° nose-up; towed from a point above its centre of gravity it flies level with a static margin of 0.27 calibres.

Derivation

Cushion thickness from the MIL-HDBK-304B minimum-acceleration curves for 2 lb/ft³ polyethylene: $t = f(G_{\text{rating}}, h, \sigma_{\text{static}})$; at $G = 150$, $h = 1.524$ m: $t \approx 41$ mm. Envelope: $L_{\text{sensor}} + 2t + 2t_{\text{wall}} + \text{rigging} < 381$ mm on every face that can be dropped.

What would change it

TC-02 drops showing the potted electronics or LED windows failing below 150 g, which would force thicker foam and a lighter maximum declaration.

---

S-03 / EVIDENCE ESTIMATE / ENABLES DECISION P-01

# S-03

## One container and adjustable simulators serve every declaration

Recommendation

**One container with re-cuttable foam pads, built for the top of the declared range; simulators with adjustable internal ballast.**

Why, plainly

Choosing the declared mass in April only helps if the hardware can follow the choice. The container is built once, for the heaviest sensor we might declare; its pads are re-cut at tech for the mass actually declared, so a lighter sensor is not over-cushioned. Simulators must match the container plus the declared sensor within 28 g, so they carry adjustable ballast.

The engineering

One container serves declarations from 3 to 8 kg with pads re-cut at tech (3 to 8 kg, ESTIMATE). Container mass is 1.00 kg built for 6 kg and 1.18 kg for 8 kg. Container mass counts toward the Mission 2 payload, so it is not wasted mass in the mission it is carried. Simulator tolerance is ±0.0283 kg.

What would change it

TC-02 showing that re-cut pads change the drop response beyond the sensor's rating.

---

S-04 / EVIDENCE COMPUTED / TC-05

# S-04

## Recover the sensor fast enough to keep the seventh lap

Recommendation

**Closed-loop drum with a slip ring and a docking latch (DB-01 option A), driven by a high-torque servo drive sized for the declared maximum; Mission 3 ballast chosen per attempt so the stow fits the window.**

Why, plainly

Mission 3 counts whole laps inside five minutes, and the sensor must be back inside before landing. At about 39-second laps, seven laps fit only if take-off plus stowing takes less than about 26 seconds. A heavier sensor winds in more slowly, so the drum drive, the ballast flown and the lap time are one decision.

The engineering

Seven laps at 39.1 s need a stow of no more than 18.3 s. The servo drum stows 3.4 kg in 14.0 s and 6 kg in 19.0 s; the brushed gearmotor of DB-01 takes 11.9 and 19.6 s;

a geared brushless drive stows any mass in about 3 s but weighs 0.70 kg, flown in every mission. Tow tension in a 4 g pull-up reaches 392 N with a 10 kg sensor, so the line and its anchor are sized for that case.

Derivation

$\text{Laps} = \lfloor (300 - t_{\text{take-off}} - t_{\text{stow}}) / t_{\text{lap}} \rfloor$, with the stow counted inside the window until a Q&A ruling says otherwise.

What would change it

A Q&A ruling that the stow time falls outside the window (then the drive choice is free), or TC-05 cycle counts showing the servo drum below 98 % recovery.

---

S-05 / EVIDENCE VERIFIED (RULES) / DECLARED AT THE GROUND MISSION

# S-05

## Drop from the full height

Recommendation

**Declare 1.524 m (60 in) for the Ground Mission.**

Why, plainly

The Ground Mission score is the sensor mass times the drop height. Every inch below the maximum throws away score that the container was designed to protect.

The engineering

The container of card S-03 is sized for the full 1.524 m drop at 150 g; at 6 kg the product reaches 9.14 kg m, the expected-field anchor.

What would change it

TC-02 failures at 1.524 m; then the declared height drops in whole inches to the highest height that passes.

---

S-06 / EVIDENCE VERIFIED AND COMPUTED / UIUC, XFOIL

# S-06

## Keep the proven airfoil

Recommendation

**SD7037 throughout the wing.**

Why, plainly

The airfoil has to work well across the racing range, take a flap well, and not lose lift when its surface is imperfect, because a student-built wing will be imperfect. SD7037 does all three, and it has been measured in a wind tunnel at our speeds.

The engineering

SD7037's maximum lift is unchanged by a leading-edge trip in the UIUC tests, while SD7062 loses 24 to 27 %. It has the best or equal-best lift-to-drag ratio of thirteen candidates across the racing lift range. Measured clean maximum lift is 1.22 to 1.30 at $Re = 2\text{–}3 \times 10^5$.

What would change it

A section with tunnel data showing equal roughness tolerance and higher flapped lift at $Re = 4\text{–}9 \times 10^5$.

S-07 / EVIDENCE ESTIMATE / TC-03

# S-07

## Lights the judges can see in the Tucson sun

Recommendation

**Three LED arrays of about 220 cd each, flash at 1 to 2 Hz, one-button sequencing on the observer's transmitter.**

Why, plainly

Mission 3 fails if the judges cannot tell OFF, SOLID and FLASHING apart from the ground in bright sun. The sequence must also be impossible to fumble mid-lap, so one button steps it.

The engineering

Required intensity per light is estimated at 223 cd for 150 to 250 m slant range; energy per Mission 3 is 1.40 Wh from the sensor's own labelled pack.

What would change it

TC-03 visibility scores below 90 % at 200 m.

# 04

# Variant A

The recommended aircraft: a compact wing with Fowler flaps that lands a heavy, late-declared sensor slowly enough to finish.

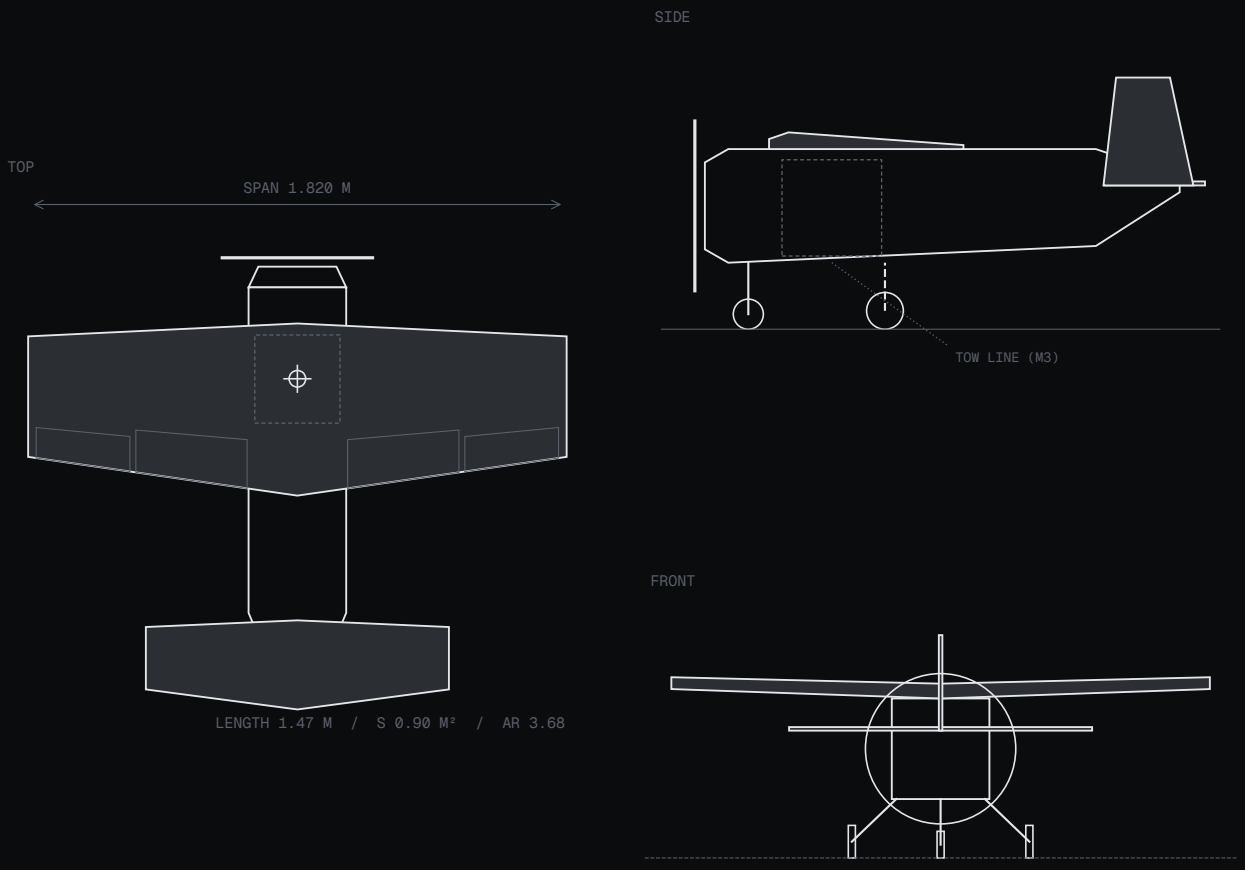
IN THIS CHAPTER

06 Specification
06 Decision cards
06 Stability and balance
06 Missions, score and risks

# A Fowler-flap compact carrier

Straight 0.9 m² wing with Fowler flaps, high wing, conventional tail, fixed tricycle gear, single 8S tractor; sensor declared at tech between 3 and 6 kg.

GENERAL ARRANGEMENT / TOP, SIDE, FRONT, ISO




SPAN 1.820 M  WING AREA 0.90 M²  PAYLOAD BAY (DASHED) CENTRED ON THE CG  CROSSHAIR: DESIGN CG, ALL LOADINGS

## PRINCIPAL CHARACTERISTICS

| SPAN | WING AREA | ASPECT RATIO | LENGTH |
|---|---|---|---|
| 1.820 M | 0.9 M² | 3.68 | 1.47 M |
| **EMPTY MASS** | **M2 GROSS** | **STALL AT M2** | **DECLARED SENSOR** |
| 6.74 KG | 13.6 KG | 12.2 M/S | 5.72 KG |
| **M2 SCORE** | **M3 SCORE** | **GM SCORE** | **STATIC MARGIN** |
| 1.44 | 2.87 | 1.44 | 16 % |
| **MEAN TOTAL** | **P10 TOTAL** | **P(TOP OF FIELD)** | **P(1ST), REPORT 90** |
| 6.76 / 7.5 | 6.30 | 0.286 | 0.348 |

## CONFIGURATION

| | | | |
|---|---|---|---|
| WING | Straight, taper 0.7, SD7037, 2° washout | SENSOR | Tungsten dart, 150 g, 3–6 kg |
| HIGH LIFT | Fowler, 30 % chord, 60 % semispan | CONTAINER | 298 x 288 x 288 mm, 1.00 kg |
| POSITION | High wing, 2° dihedral | D&R | Closed-loop drum, slip ring |
| TAIL | Conventional, $V_h$ 0.55, $V_v$ 0.040 | BAY AND EXIT | Both on the CG |
| GEAR | Fixed tricycle, faired | STRUCTURE | Rib and spar, CF tube spar |
| PROPULSION | 1 x AT7215 KV270, APC 20x15E | JOINT | Two-piece wing |
| BATTERY | 8S 3300 mAh, 97.7 Wh | RADIO | ExpressLRS 2.4 GHz, PWM |
| FUSE | 80 A blade, 58 V rated | FAIL-SAFE | Set per output (§4.2.5) |

## MISSION PLAN

| | | |
|---|---|---|
| M1 | Waived; the first successful Mission 2 credits it. | 1.00 |
| M2 | One unit at the declared mass; simulators only if the weigh-in leaves margin. | 1.44 |
| M3 | Ballast chosen per attempt for laps and a safe landing; stow inside the window. | 2.87 |
| GM | Declared sensor from 1.524 m, six drops. | 1.44 |

08

# Variant A: the Fowler-flap compact carrier

**What it is**

8.1

Variant A is a 0.9 m² straight wing on the full 1.82 m span, with Fowler flaps across the inner 60 % of each half-span, a high-mounted wing over a box fuselage, a conventional tail, fixed tricycle gear and a single 8S tractor. It carries one dense sensor whose maximum mass is declared at technical inspection between 3 and 6 kg, and it is laid out so that the payload bay and the tow-line exit both sit exactly on the centre of gravity.

Why A

**Of every design that passed the gates, A has the highest probability of posting the top mission total in the simulated field, 0.286 against 0.128 for the DB-01 control, and the smallest worst-case regret across the five stress scenarios (0.0268).**

A-01 / EVIDENCE ESTIMATE (FLAP DATA) AND COMPUTED / FREEZE 19 FEB 2027

A-01

## A small wing that lands like a big one

Recommendation

**0.9 m² with Fowler flaps (trimmed landing $C_{L,\max}$ about 1.9).**

Why, plainly

A bigger wing lands a heavy load more slowly, but it is heavier and draggier in every lap. A Fowler flap gives the small wing the landing speed of a big one: it slides back and down, adding area and lift only when it is needed. The result is the slowest landing of any design we scored at this payload, and a slow landing is what makes Mission 2 and Mission 3 finish.

The engineering

At Mission 2 weight (median 13.6 kg) Variant A stalls at 12.2 m/s, against 13.5 m/s for the DB-01 control with an honest plain flap. Per-attempt success rises to 0.533 in Mission 2 and 0.516 in Mission 3. The trimmed landing $C_{L,\max}$ is 1.92 (range 1.76–2.05), from Platt's NACA TR 534 Fowler data through the TN 663-validated area method. That trim figure assumed a tail arm of three mean chords. A's tail arm is 0.943 m, about 1.9 chords, which costs about 0.06 more in trimmed lift. The *Fowler underdelivers* scenario takes 8 % off $C_{L,\max}$, more than covering that loss.

Derivation

$V_s = \sqrt{\dfrac{2mg}{\rho S C_{L,\max}}}$; at $m = 13.6$ kg, $S = 0.9$ m², $\rho = 1.064$ kg/m³, $C_{L,\max} = 1.92$: $V_s = \sqrt{\dfrac{2 \times 13.6 \times 9.807}{1.064 \times 0.9 \times 1.92}} \approx 12.0$ m/s.

What would change it

A bench and flight test of the Fowler flap below 1.75 in trimmed $C_{L,\max}$, or a track mechanism that jams in more than one cycle in a hundred. Either moves the team to Variant B (Section 10).

A-02 / EVIDENCE COMPUTED (AVL 3.52) / CDR LAYOUT

# A-02

## Put the payload and the tow line on the centre of gravity

Recommendation

**Wing placed so the empty aircraft balances at the design CG; payload bay and tow exit centred on it.**

Why, plainly

The aircraft flies four very different loadings: empty, carrying a 7 kg unit, towing a 3 kg sensor, towing a 6 kg sensor. If any of them moves the balance point, the pilot meets a different aircraft on each flight, and a hand-flown aircraft with no autopilot is only as good as its consistency. Putting the heavy items on the balance point makes all four feel the same.

The engineering

AVL places the wing-and-tail neutral point at 0.489 m aft of the propeller plane; the design CG is at 0.409 m, a 16 % margin. Because the wing is positioned so the empty aircraft balances there, every loading stays between 16.0 and 16.2 % of mean chord. The fuselage removes about 4 % (Raymer's fuselage term, an estimate), leaving about 12 % in flight. AVL gives $C_{m\alpha}$ = -1.14, $C_{n\beta}$ = 0.0817 and $C_{l\beta}$ = -0.0154 per radian: stable in pitch, in weathercock and in roll.

Derivation

Static margin = $(x_{np} - x_{cg})/\bar{c}$. Horizontal tail: $S_h = V_h S\bar{c}/l_t$ with $V_h$ = 0.55, raised above the usual 0.5 because a wing of aspect ratio 3.7 throws strong downwash on the tail.

What would change it

Measured CG at the first weigh-in more than 3 % of mean chord from the design point; ballast or battery position then trims it.

A-03 / EVIDENCE VERIFIED (FUSES) AND COMPUTED / TC-06

# A-03

## One proven motor, and a fuse that is actually rated for the pack

Recommendation

**Single AT7215 KV270 with an APC 20x15E on an 8S 3300 mAh pack; an 80 A blade fuse rated for 58 V.**

Why, plainly

The DB-01 propulsion study already showed this motor, propeller and pack to be the best single system, and twin layouts lose. What it missed is the fuse. The rules demand a blade fuse, and the common 100 A blade fuse is rated for only 32 V, while a full 8S pack sits at 33.6 V. A fuse used above its voltage rating may not clear a fault safely, so we pick the strongest blade fuse that is genuinely rated for the pack.

The engineering

Littelfuse rates its 100 A MAXI blade at 32.0 V DC; its 58 V MAXI series stops at 80.0 A. The 80 A part is legal (the cap is the lower of 100 A and the pack's continuous rating) and carries the cruise and turn currents with margin; take-off peaks of about 93 A are inside a slow-blow fuse's time-current curve, which TC-06 must confirm. Static thrust is 101 N. Above about 17.5 kg of Mission 2 gross the 80 %-usable energy budget, not thrust, becomes the limit.

What would change it

A 58 V, 100 A blade fuse in stock (the ESKA part is listed only on request), or TC-06 showing the 80 A fuse opening in a simulated Mission 2.

---

A-04 / EVIDENCE COMPUTED AND FIELD PRACTICE / FREEZE 19 FEB 2027

# A-04

## High wing, conventional tail, fixed tricycle gear

Recommendation

**Keep DB-01's layout choices; fair the gear; tall nose leg for the 20-inch propeller.**

Why, plainly

None of these choices wins points on its own; each one avoids losing them. A high wing keeps the wing clear of the tow line leaving the belly and puts the bay under the spar. A conventional tail is the only common tail that can give the rule-mandated fail-safe of full up elevator with full right rudder. Fixed gear avoids the gear-up landings that retractable gear invites, for about one hundredth of a point of drag.

The engineering

The 20-inch propeller needs its axis 0.369 m above the runway for tip clearance, so the nose leg is tall and the gear is faired. Retractable gear pays only below 120 to 240 g of mechanism (DB-01). The field gives no configuration a measurable edge (Fisher tests, $p \geq 0.24$), so these follow from physics, packaging and risk.

What would change it

The pilot's strong preference for a taildragger after test flying, which would also shorten the nose leg.

Table 05

Variant A's main risks, each with the test or decision that retires it.

| Risk | L | C | Mitigation | Retired by |
|---|---|---|---|---|
| Fowler tracks bind or misalign, so the flaps do not fully extend | M | H | Printed carriages on carbon rods; single servo per side with travel stops | Bench cycles, 100 at full load |
| Airframe heavier than the fit | M | M | Declare at tech after weigh-in; A holds the top rank at +15 % mass | Weigh every part |
| Landing a 13.6 kg aircraft at about 12 m/s on pavement | M | H | Long flap travel; hot-day declaration rule | Heavy-landing flight tests |
| A 6 kg dart swings in turns (0.5 Hz pitch mode, lightly damped) | M | H | Tow eye above the sensor's CG; small fins far aft | TC-01 at full weight |
| Stow takes 19 s at 6 kg and can cost the seventh lap | M | M | Ballast chosen per attempt | TC-05 timing |
| Fuse opens on take-off | L | H | 58 V slow-blow part | TC-06 fuse duty |

### What can go wrong

8.2

# 05

# Variant B

The hedge: a broader wing with simple slotted flaps that wins exactly where Variant A is weakest, and shares everything else.

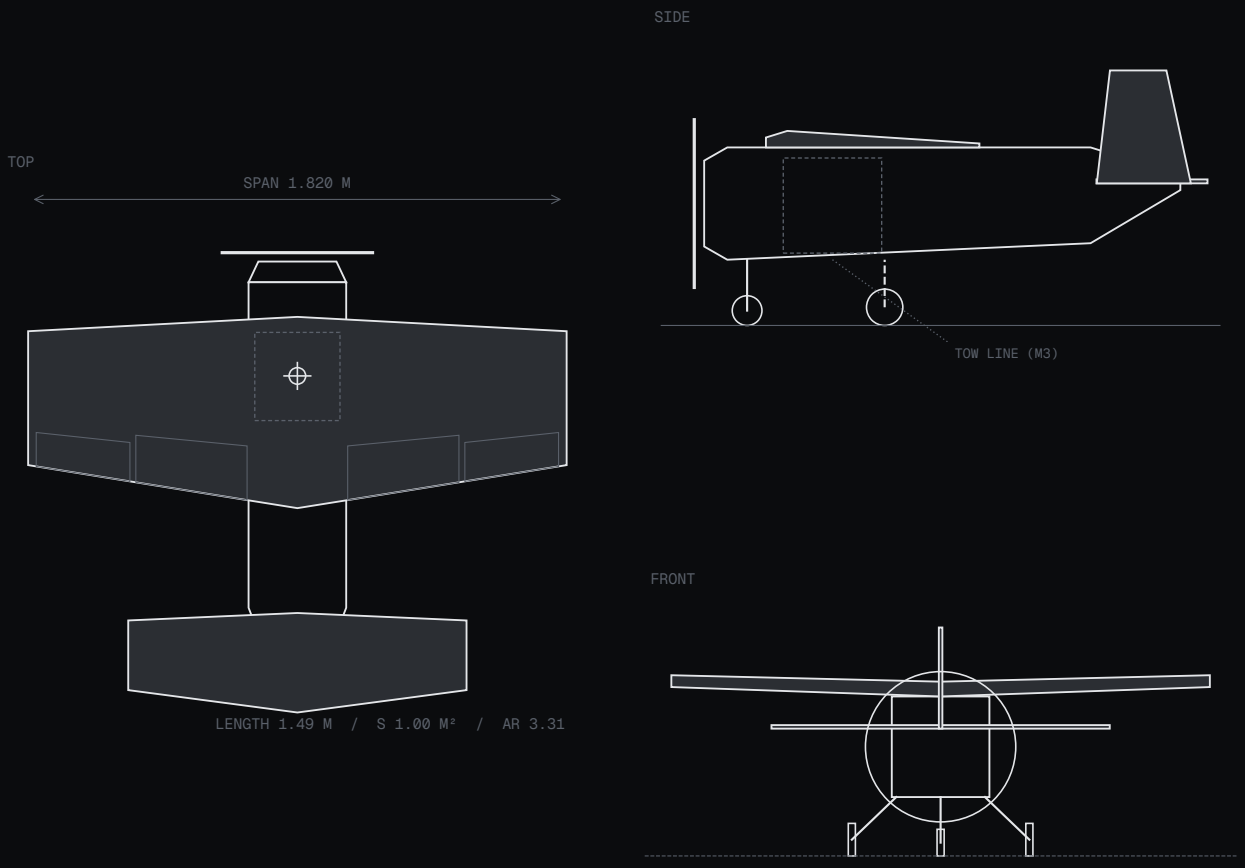
IN THIS CHAPTER

07 Specification
07 How B differs, and why
07 What it shares with A

# B Slotted-flap broad-wing carrier

Straight 1.0 m² wing with single-slotted flaps; otherwise the same fuselage, tail type, gear, propulsion and payload as Variant A.

GENERAL ARRANGEMENT / TOP, SIDE, FRONT, ISO




SPAN 1.820 M  WING AREA 1.00 M²  PAYLOAD BAY (DASHED) CENTRED ON THE CG  CROSSHAIR: DESIGN CG, ALL LOADINGS

## PRINCIPAL CHARACTERISTICS

| SPAN | WING AREA | ASPECT RATIO | LENGTH |
|---|---|---|---|
| 1.820 M | 1.00 M² | 3.31 | 1.49 M |
| **EMPTY MASS** | **M2 GROSS** | **STALL AT M2** | **DECLARED SENSOR** |
| 7.12 KG | 13.7 KG | 12.9 M/S | 5.42 KG |
| **M2 SCORE** | **M3 SCORE** | **GM SCORE** | **STATIC MARGIN** |
| 1.40 | 2.82 | 1.40 | 16 % |
| **MEAN TOTAL** | **P10 TOTAL** | **P(TOP OF FIELD)** | **P(1ST), REPORT 90** |
| 6.62 / 7.5 | 6.13 | 0.238 | 0.298 |

## CONFIGURATION

| | |
|---|---|
| WING | Straight, taper 0.7, SD7037, 2° washout |
| HIGH LIFT | Single-slotted, 30 % chord, 60 % semi-span |
| POSITION | High wing, 2° dihedral |
| TAIL | Conventional, $V_h$ 0.55, $V_v$ 0.040 |
| GEAR | Fixed tricycle, faired |
| PROPULSION | 1 x AT7215 KV270, APC 20x15E |
| BATTERY | 8S 3300 mAh, 97.7 Wh |
| FUSE | 80 A blade, 58 V rated |

| | |
|---|---|
| SENSOR | Same as A |
| CONTAINER | Same as A |
| D&R | Same as A |
| BAY AND EXIT | Both on the CG |
| STRUCTURE | Rib and spar, CF tube spar |
| JOINT | Two-piece wing |
| RADIO | ExpressLRS 2.4 GHz, PWM |
| SHARED PARTS | Fuselage, tail type, gear, power, payload |

## MISSION PLAN

| | | |
|---|---|---|
| M1 | Waived; the first successful Mission 2 credits it. | 1.00 |
| M2 | One unit at the declared mass. | 1.40 |
| M3 | Ballast chosen per attempt; stow inside the window. | 2.82 |
| GM | Declared sensor from 1.524 m, six drops. | 1.40 |

09

# Variant B: the slotted-flap broad-wing carrier

**Why a second aircraft**

9.1

Variant A's lead rests on one component the team has never built: a Fowler flap that slides on tracks. If that flap gives less lift than the tunnel data promise, or proves unreliable, A's advantage disappears. Variant B is the aircraft that wins in exactly that case. It replaces the Fowler flap with a single-slotted flap, which hinges like an ordinary flap and opens a slot, and buys back the lost lift with 10 % more wing area.

Selection

**The rule fixed on 27 October asked for a B that differs from A in a feature the rules freeze at the design report. The only such candidates the search could score were twin-motor layouts, and neither covers A's weakness: the Fowler-flap twin shares it and falls furthest when the flap underdelivers, and the slotted-flap twin scores below B in all five scenarios (Figure 02). B is therefore the best candidate in A's worst scenarios, as the rule's second clause asks, and it differs from A in wing area and flap type. The rules' list of major configuration changes does not include either, though the Tech Inspector has the final word.**

B-01 / EVIDENCE ESTIMATE (FLAP DATA) AND COMPUTED / SWITCH GATE 15 JAN 2027

B-01

## Trade the sliding flap for a hinged one and a larger wing

Recommendation

**1.0 m² wing with single-slotted flaps on the same span.**

Why, plainly

A slotted flap is a hinge and a gap: easy to build from laser-cut ply and printed brackets, easy to inspect, hard to jam. It gives less extra lift than a Fowler flap, so the wing grows by a tenth to land at nearly the same speed. The bigger wing costs some speed in every lap, which is why B scores a little lower than A when everything works, and why it is the better aircraft if the Fowler flap disappoints.

The engineering

Trimmed landing $C_{L,\max}$ 1.62 (range 1.45–1.78) from Wenzinger and Harris's NACA TR 664 slotted-flap data. At Mission 2 weight (median 13.7 kg) B stalls at 12.9 m/s. Its probability of the top mission total is 0.238 in the base case, and 0.238 if the Fowler flap underdelivers, against 0.221 for A in that case. Its regret in A's two worst scenarios is 0.0102 and 0.005, inside the Monte Carlo uncertainty of zero.

What would change it

Fowler bench tests passing by 15 January, in which case A is built; a heavier airframe than the fit, which favours A (B's regret rises to 0.0526).

B-02 / PROGRAMME DECISION / PROCUREMENT NOW

# B-02

## Share every long-lead part with A

Recommendation

**Same fuselage, tail type, gear, propulsion system, sensor, container, drum and radio.**

Why, plainly

A hedge is only cheap if it shares the parts that take longest to buy. Every long-lead item, the motor, propeller, packs, ESC, fuse, receiver, drum drive and slip ring, is common to both aircraft, so ordering for A is ordering for B. Only the wing and its flaps differ, and those are built in house.

The engineering

B's empty mass is 7.12 kg against 6.74 kg for A (fit and flap mechanism). Its neutral point is 0.505 m and design CG 0.417 m, with the same 16 % margin across every loading, and its horizontal tail is 0.327 m² against A's 0.262 m².

What would change it

Nothing in the evidence; this is a programme choice that keeps the switch cheap.

# 06

# Head to head

How the two aircraft compare when the inputs go against them, which one to build, and the test result that switches the team from one to the other.

IN THIS CHAPTER


| | |
|---|---|
| 08 | Five stress scenarios |
| 08 | The report decides the win |
| 08 | Recommendation and switch |
| 09 | Decision calendar and playbook |
| 10 | Test gates and schedule |

# 10 Head to head

## Five ways the inputs could go wrong

10.1

Each candidate was re-scored in five scenarios built to attack Variant A: the base case; the Fowler flap delivering 8 % less lift with double the mechanism hazard; every airframe 15 % heavier than the fit; every attempt 8 points less likely to succeed; and the Fowler and mass problems together. Figure 02 shows the probability of the top mission total in each, with its Monte Carlo 95 % interval.

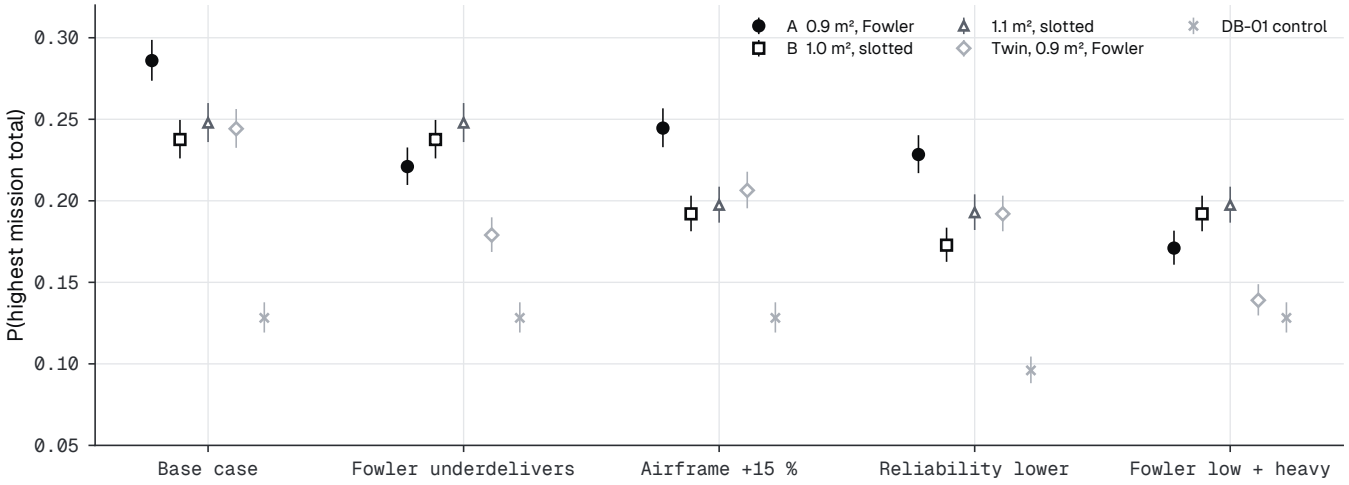



Figure 02

Variant A leads in three of the five scenarios and stays within 0.03 of the leader in the other two; the slotted-flap wings lead where the Fowler flap underdelivers, Variant B within the Monte Carlo uncertainty of the leader. The twin never leads a scenario, and the DB-01 control trails throughout. Bars: 95 % Monte Carlo intervals over 5,000 simulated fields.

Table 06

Probability of the top mission total by scenario, and each candidate's regret: how far it falls short of the best candidate in that scenario.

| Candidate | Base | Fowler low | Mass +15 % | Reliability low | Combined | Max regret |
|---|---|---|---|---|---|---|
| A: 0.9 m², Fowler | 0.286 | 0.221 | 0.245 | 0.228 | 0.171 | 0.0268 |
| B: 1.0 m², slotted | 0.238 | 0.238 | 0.192 | 0.173 | 0.192 | 0.0562 |
| 1.1 m², slotted | 0.248 | 0.248 | 0.197 | 0.193 | 0.197 | 0.0472 |
| Twin, 0.9 m², Fowler | 0.244 | 0.179 | 0.206 | 0.192 | 0.139 | 0.0688 |
| DB-01 control | 0.128 | 0.128 | 0.128 | 0.096 | 0.128 | – |

**The aircraft makes the win possible; the report decides it**

10.2

The competition score multiplies the mission total by the report score, so the same aircraft has very different chances depending on the design report. With Variant A, the probability of finishing first overall is 0.116 at a report score of 80, 0.263 at 86, 0.348 at 90 and 0.409 at 94 (Figure 03). The simulated 2027 field's best mission total has a median of 6.62 (80 % range 6.22 to 7.04), because rivals who read the rules as we do will also declare heavy sensors: the field's Ground Mission maximum has a median of 11.4 kg m.

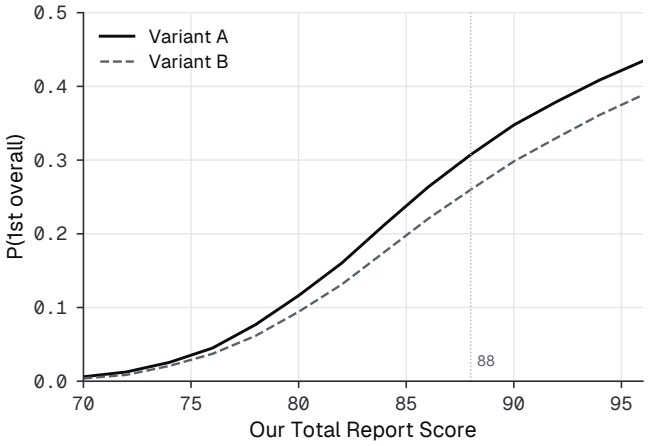



**Figure 03**

Near a report score of 88, each report point adds about two points of win probability. At a report score of 80, even Variant A wins only about one time in nine.

DATA QUALITY / FIELD MODEL

The field simulator reproduces the real winning mission totals of 2022 to 2025 inside its 10 to 90 % band, and puts 2026's 5.61 just below its band, so it errs toward stronger rivals. It reproduces the number of teams completing every mission in all five years. Its 2027 maxima are anchored on 2021 and 2025 results and drawn over a wide range. Probabilities here are comparative evidence between candidates on identical draws, not forecasts of the fly-off.

---

**Recommendation**

RECOMMENDATION / TEAM DECISION BY 15 JAN 2027

10.3

## H-01

Recommendation

## Build Variant A; keep Variant B one wing away

**Variant A as the competition aircraft; Variant B's wing as the funded fallback, decided at a dated gate.**

Why, plainly

A is the best aircraft in most of the futures we can foresee and never far from the best in the others. Its one real weakness is a mechanism the team has not built yet. B covers that weakness with parts the team already knows how to make, and because the two aircraft share everything except the wing, carrying B costs one extra wing, not a second programme.

The engineering

Base case: A 0.286, B 0.238, control 0.128 probability of the top mission total; mean totals 6.76, 6.62 and 6.33; worst-tenth totals 6.30, 6.13 and 6.00. A's largest regret across five scenarios is 0.0268; B's is 0.0562.

What would change it

**Switch to B** if, by 15 January 2027, the Fowler flap on the bench or on the prototype wing gives less than 1.75 in trimmed $C_{L,\max}$, or jams in more than one of a hundred full-load cycles. Otherwise build A and stop work on B's wing.

## 11 Decision calendar and mission-day playbook

### Decide each number when it can be decided well

11.1

The largest single gain in this dossier came from timing, not hardware: choosing the declared sensor mass in April instead of now. The same discipline applies to every number the team must fix.

Table 07

When each decision is made, and what is known at that moment.

| When | Decide | On the basis of |
|---|---|---|
| Now to 15 Dec 2026 | Order long-lead parts common to A and B; build the Fowler test section | DB-02; PR-01 |
| 15 Jan 2027 | A or B (card H-01) | Fowler bench and prototype results |
| 19 Feb 2027 | Configuration freeze: wing, tail, gear, motor count | First flights, weigh-in |
| Technical inspection | Declared sensor mass (3 to 6 kg), simulator count, wingtip-test mass | Weighed aircraft, measured lap times, a hot-day plan ($\rho = 1.04$ kg/m³) |
| Each Mission 3 attempt | Ballast flown, propeller | Wind, density, lap times so far |
| Ground Mission | Drop height in whole inches (1.524 m unless TC-02 failed) | TC-02 results |

### Mission-day playbook

11.2

- **Mission 1.** Waive it at the flight line. The first successful Mission 2 scores it.
- **Mission 2.** Fly one unit at the declared mass. Add a simulator only if the weigh-in left margin and the afternoon is cool; the hot-day plan already sets the declared mass so that one unit is safe on the hottest likely afternoon.
- **Mission 3.** Deploy before the first downwind turn with lights off, step the sequence with one button after laps one and two, and choose the ballast so the stow fits inside the window: a lighter dart that keeps the seventh lap beats a heavier one that loses it.
- **Ground Mission.** Declare 1.524 m. Re-cut pads only at technical inspection, never on the day.
- **Wind.** Above about 8 m/s sustained (the 90th percentile at TIMPA), fly the lighter Mission 3 ballast and land with full flap into wind.

# 12 Test gates and schedule

### Tests that retire the decisions

12.1

Every recommendation in this dossier names the result that would overturn it. Table 08 collects those results into test gates, each with a pass criterion and the decision it unlocks. TC-01 to TC-06 already exist as test cards; TC-07 to TC-09 are new.

Table 08

Test gates. Each pass criterion comes from a decision card; failing a gate triggers that card's alternative.

| Card | Test | Pass criterion | Unlocks |
|---|---|---|---|
| TC-01 | Tow stability, eye above CG, 3 and 6 kg dart | No sustained swing or spin at 20 to 30 m/s | Sensor design (S-02) |
| TC-02 | Six drops at 1.524 m, 150 g sensor, pads cut for 3 and 6 kg | No damage, lights work after every drop | Declared range, drop height (S-02, S-05) |
| TC-03 | Light visibility in sun | Modes identified at 200 m, 90 % or better | Light design (S-07) |
| TC-04 | Five-minute drills | Mission 2 loading and Mission 3 attach in under 240 s | Bay and latch design |
| TC-05 | Drum cycles at 3, 4.4 and 6 kg | 98 % recovery; stow time logged | D&R drive (S-04) |
| TC-06 | Propulsion bench and fuse duty | Static thrust 95 N or more; 80 A fuse survives a Mission 2 current profile | Propulsion and fuse (A-03) |
| TC-07 | Fowler section bench test (new) | Trimmed-equivalent $C_{L,\max}$ 1.75 or more; under 1 jam in 100 cycles | A or B (H-01) |
| TC-08 | First weigh-in and CG check (new) | Empty mass within 15 % of the fit; CG within 3 % of mean chord | Declared-mass plan |
| TC-09 | Lap timing with GPS logger (new) | Laps within 15 % of the model | Lap factor, Mission 3 ballast plan |

### Schedule

12.2

- **November to mid-December 2026.** Order long-lead parts; build the Fowler test section and run TC-07; run TC-01, TC-02, TC-05 and TC-06 on the bench.
- **15 January 2027.** A-or-B gate (card H-01).
- **January to mid-February.** Build and fly the competition airframe (one airframe only may be presented at the contest, so a simpler practice airframe flies first); TC-08 and TC-09; proof-of-flight video.
- **19 February.** Design report and configuration drawing; configuration frozen.
- **March to April.** Pilot hours on type, five-minute drills, hot-day performance checks; technical inspection and the declared-mass decision at the fly-off.

# 07

# Questions, terms and sources

The rules questions worth asking in round one, the vocabulary of this dossier, and every source it rests on.

IN THIS CHAPTER

10 Rules Q&A, ranked by points at stake
11 Glossary
12 Sources

# 13 Rules Q&A, ranked by points at stake

### How to use this list

13.1

Round one of the official Q&A opens on 1 November and accepts five questions; later rounds accept three. Each question below states what rides on the answer and the design the team builds to until it is answered, so no answer can strand the aircraft.

Table 09

Round-one questions, in order of the points they protect. The fallback is what the design assumes until the answer arrives.

| Rank | Question, as it would be submitted | What rides on it | Fallback until answered |
|---|---|---|---|
| 1 | In Mission 3, is the time to stow the sensor after the final lap inside the 300 s flight window? | Seven laps against six at 39 s laps; the choice of drum drive | Assume it counts; trim the Mission 3 ballast so the stow fits |
| 2 | Staging allows no adjustments such as opening hatches. May a hatch be opened and latched inside the box, without tools, to load payload and batteries? | A misreading fails staging | Enter with the bay open; latch-only closure inside the box |
| 3 | Does the 381 mm ring-to-bottom limit apply to a sling to the container's corners, and may the sling legs sit at 10 to 15°? | The container envelope and the declared drop height | 150 g sensor and 41 mm foam fit at 15°; a lower drop height costs 8 % of the Ground Mission bonus per 5 in |
| 4 | Do identical bolt-on spare panels and control surfaces count as repair of the original airframe? | Whether one broken surface ends the contest | Carry identical bolt-on spares as repair stock |
| 5 | May the sensor's internal battery stay in the sensor while it is packed in its container for the Ground Mission? | Ground Mission compliance | A removable sensor battery fitted at the light check |
| Later | Is the 1.5-span deployment distance checked statically or in flight? | Line length | Static exit-to-nose length with a 0.10 m margin |
| Later | Is wingspan the projected tip-to-tip width when vertical tip surfaces are fitted? | Only matters if winglets return | Not needed: winglets were eliminated (Table 04) |

Already answered

**Packs may change between flights, within the identical-pack rule (FAQ, layout lines 1670–1673). Rate gyros and stabilisation are allowed (line 600). Neither needs a question.**

# 14 Glossary

# 15 Sources

### How sources are graded

15.1

Each row gives the source, what was taken from it, when it was read, and its nature: measured data, a specification, a computation, a rule, or a method. Sources used by DB-01 and the analysis scripts are indexed in the repository's `SOURCES.md`.

Table 10

Terms used in this dossier.

| Term | Meaning |
|---|---|
| Declared sensor mass | The maximum sensor mass the team states at technical inspection; it fixes every Mission 2 unit and the Ground Mission score |
| Payload unit | One container holding the sensor at its declared mass, or one simulator matching it within 28 g |
| Field maximum | The best result any team posts in a mission, our own included; every bonus is a ratio to it |
| P(top) | Probability that our total mission score is the highest in the field |
| P(1st) | Probability that our competition score (report times mission, plus participation) is the highest |
| P10, P90 | The score below which the worst tenth, or above which the best tenth, of the uncertain outcomes fall |
| Paired draws | The same 120 sets of uncertain inputs applied to every candidate, so differences are real |
| Trimmed $C_{L,\max}$ | Maximum lift after the tail's balancing load is subtracted |
| Oswald *e* | How close the wing's drag due to lift is to the ideal; 1.0 is ideal |
| Fowler flap | A flap that slides aft on tracks as it lowers, adding wing area and opening a slot |
| Single-slotted flap | A flap that rotates down and opens a gap that re-energises the flow over it |
| Stall speed $V_s$ | The slowest speed at which the wing can carry the aircraft's weight; landings are made somewhat above it |
| Decision timing | Choosing each number at the moment the information to choose it well exists, and not before |

| SOURCE | REFERENCE AND USE | NATURE |
|---|---|---|
| REF | AIAA (2026). *2026-27 Design/Build/Fly Competition Rules (draft of 2 September 2026).* rules/2027_DBF_Rules_Draft.pdf; SHA-256 d94773fb…ae1<br>All rule quotes, by line of the layout text | PRIMARY |
| REF | AIAA (2026). *DBF competition summaries and top-3 design reports, 2021-26.* aiaa.org; registry analysis/data/sources.toml (1,008 verified values)<br>Field data, mass, drag, lap times | PRIMARY |
| REF | Team (DB-01) (2026). *DB-01 Sizing and configuration decision brief, revision A, 14 October 2026.* decisions/DB-01_sizing_and_configuration.pdf<br>Baseline and control design | PRIMARY |
| REF | Team (analysis) (2026). *Analysis Scripts 1-6 and 3b; SOURCES.md registry (113 sources).* analysis/; SOURCES.md<br>Scoring, sizing, propulsion and Monte Carlo models | PRIMARY |
| S1-RULES-2026-09-02 | AIAA (2026). *2026-27 DBF rules, DRAFT dated 2 Sep 2026.* REPO/rules/2027_DBF_Rules_Draft.pdf, SHA-256 d94773fb17ee3bdac88191bf4cbce96c8d0b2166b9b089ed670052bf9ca3bae1; read 2026-10-27 and 2026-10-29<br>all 99 clause quotes in out/rules/compliance.csv, checked against analysis/data/raw/aiaa-dbf-2027-rules-draft.layout.txt by tests/test_v_rules.py | RULE |

| SOURCE | REFERENCE AND USE | NATURE |
|---|---|---|
| S1-RESULTS-2022-26 | AIAA, parsed in REPO/analysis/data/dbf_results.csv (2026). *AIAA DBF competition results 2022-2026 (513 team rows).* REPO/analysis/data/dbf_results.csv; read 2026-10-29<br>report and mission score distributions, winning scores 501.26-567.51, participation 3 | MEASURED |
| S1-ATTEMPTS | REPO/analysis/data/attempts.py (AIAA competition summaries 2022-26) (2026). *Attempt-budget model, 2.4-2.9 official attempts per team that passed tech, 57-73 % of attempts scored.* REPO/analysis/data/attempts.py; read 2026-10-29<br>p_complete for the M1 waiver value | COMPUTED |
| S1-ADAPTER | S12 (this effort) (2026). *S12 integration adapter over DB-01 s6_studies.evaluate.* W/src/integration/adapter.py; read 2026-10-29<br>H1 study: 120 paired draws seed 2027, expected field; REF-D01 6.3326 | COMPUTED |
| S3-naca-tr534 | R. C. Platt (1935). *Aerodynamic characteristics of a wing with Fowler flaps, including flap loads, downwash, and calculated effect on take-off.* NACA Report 534 (NTRS); local copy out/aero_mono/raw/naca/tr534.pdf; read 2026-10-28<br>fig. 5 (p. 4): CLmax of a Clark Y wing AR 6, Re 6.1e5, with 0.20c/0.30c/0.40c full-span Fowler flaps, 1.29 clean to 2.45/2.83/3.17 on retracted area; Tables II-III (p. 15): Cm c/4 -0.085 clean, -0.508 at alpha 0 with 0.20c flap at 30 deg | MEASURED |
| S3-naca-tr664 | C. J. Wenzinger, T. A. Harris (1939). *Wind-tunnel investigation of an N.A.C.A. 23012 airfoil with various arrangements of slotted flaps.* NACA Report 664 (NTRS); local copy out/aero_mono/raw/naca/tr664.pdf; read 2026-10-28<br>fig. 30: increments of section Clmax for split, plain, external-airfoil, Fowler and slotted 2-h flaps vs deflection (read by eye, +/-0.05); p. 29: slotted-flap pitching moment equals plain-flap moment at equal lift | MEASURED |
| S3-leng2022 | Leng et al. (2022). *A study of high-lift devices for low Reynolds number applications.* ICAS 2022; local copy out/aero_mono/raw/icas2022/leng2022.pdf; read 2026-10-28<br>Sec. 6.3-6.4: DAE31, Re 2.3e5, clean CLmax 1.48; single-slotted 30 % flap at 30 deg CLmax 2.5 (+1.0); ESDU estimate +0.587; DATCOM 1.10 | MEASURED (quasi-2-D tunnel with end plates) |
| S3-williamson2012 | G. Williamson (2012). *Plain flaps on low-Re airfoils (UIUC MS thesis); UIUC LSAT Vol 6.* REPO analysis/data/raw/aero/williamson-2012-thesis.pdf and uiuc-lsat-v6-data.zip; read 2026-10-29<br>AG40d-02r 25 % plain flap: dCl, dCm, dClmax vs deflection 5-40 deg at Re 3e5; flap drag files | MEASURED |
| S3-uiuc-lsat-sd7037 | Selig et al. (1995). *UIUC Low-Speed Airfoil Tests Vol 1-3, SD7037 drag files (DRAG01-03.TXT).* REPO analysis/data/raw/aero/uiuc-lsat-v1-data.zip (also v2, v3); read 2026-10-29<br>SD7037 cd at Re about 3e5: clean cdmin 0.0071-0.0077; 0.0092-0.0097 at cl 0.7; 0.0131-0.0145 at cl 1.0; tripped 0.0125 / 0.0178 / 0.0233 | MEASURED |

| SOURCE | REFERENCE AND USE | NATURE |
|---|---|---|
| S3-xfoil | computed with REPO analysis/tools/xfoil.exe via analysis/xfoil_tools.py (import only) (2026). *XFoil 6.99 polars: SD7037 Re 3e5-1.1e6 Ncrit 9 and 5; SD7034, E387, MH114 Re 5e5 and 8e5.* ; read 2026-10-29<br>profile drag a + kp CL^2; section comparison; XFoil CLmax over-predicts (SD7037 Re 3e5: 1.38 vs 1.22-1.30 measured) | COMPUTED |
| S3-avl | M. Drela, H. Youngren (2025). *AVL 3.52 vortex lattice: Trefftz-plane e, critical-section loading, winglet, flap-twist and biplane studies*. ; read 2026-10-29<br>e of tapered wings 0.988-0.999; winglet de; biplane k ratio 0.77 at gap/span 0.15 (literature about 0.78, Prandtl 1924, not re-read) | COMPUTED |
| S3-kroo2000 | I. Kroo (2001). *Drag due to lift: concepts for prediction and reduction*. Annu. Rev. Fluid Mech. 33, 587-617; read not re-read this session; method relayed by the orchestrator 29 Oct 2026<br>fuselage factor 1 - 2 (d/b)^2; K_visc about 0.38 CD0 | METHOD |
| S3-hoerner | S. F. Hoerner (1965). *Fluid-Dynamic Drag*. ; read not re-read; standard formulas<br>fuselage form factor 1 + 1.5 (d/l)^1.5 + 7 (d/l)^3 | METHOD |
| S3-tn663-cal | REPO analysis/wing_clmax_validation.py (2026). *NACA TN 663 area-method calibration, kappa 1.086 (clean) and 1.075 (flap increment)*. ; read 2026-10-29<br>area method calibration and range | COMPUTED |
| S3-gear | REPO analysis/out/config_trades_report.txt; claude_docs/evidence_plan.md (E06) (2026). *GT 2026 measured fixed-gear drag area 9.9 cm2, Joanneum 2022 predicted 19.4 cm2; NACA TR 485 fairing cuts drag about 23 %*. ; read 2026-10-29<br>gear drag area range 9.9-19.4 cm2 | MEASURED / COMPUTED |
| S3-rq4-cfd | S10 / REPO cfd/cases/rq4_mock (2026). *rq4_mock CFD RESULTS.md (indicative only)*. ; read 2026-10-29<br>non-wing drag area about 0.0070 m2 at alpha 0 as a cross-check, never a calibration | COMPUTED |
| S8-MILHDBK304B | U.S. Department of Defense (data by AFPEA, Wright-Patterson AFB) (1978). *MIL-HDBK-304B, Package Cushioning Design, Appendix V Graphs 10.36, 10.42, 10.48 (polyethylene 2 pcf)*. https://everyspec.com/MIL-HDBK/MIL-HDBK-0300-0499/MIL_HDBK_304B_1778/; read 2026-10-16 (minima, REPO analysis/data/cushion_pe2.toml); 2026-10-29 (curve shape, page 251 re-rendered)<br>minimum peak g per thickness at 36-48 in drops (verified); normalised curve shape g/gmin versus stress/stress_opt read by eye from curves 3 and 4 (estimate, +-10 %) | MEASURED |
| S8-RULES | AIAA (2026). *2026-27 DBF rules draft, 2 Sep 2026, sections 3.1.1-3.1.3, 3.3.4, Figure 1*. REPO/rules/2027_DBF_Rules_Draft.pdf (layout text analysis/data/raw/aiaa-dbf-2027-rules-draft.layout.txt); read 2026-10-29<br>L438-453 sensor, L538-542 ring-to-bottom 15 in, L556-560 simulators, L996-1008 Ground Mission, L1002 whole inches | RULE |

| SOURCE | REFERENCE AND USE | NATURE |
|---|---|---|
| S8-ASTM-B777 | ASTM International; GS Alloy (2026). *ASTM B777-15 (reapproved 2026), Tungsten Base High-Density Metal; GS Alloy datasheet*. https://gsalloy.com/product/astm-b777-tungsten-high-density-metal/; read 2026-10-29<br>density classes 1-4: 16.85-17.25, 17.15-17.85, 17.75-18.35, 18.25-18.85 g/cc; design value 17.0 g/cc | SPEC |
| S8-TUNGSTEN-RETAIL | tungsten.com (reseller) (2026). *tungsten.com 1 lb bags of tungsten alloy shot and cubes, 17.6 g/cc, US$499.99-799.99*. https://shop.tungsten.com/tungsten-alloy-cubes-various-mm-sizes-industrial-use-1-lb-bags/; read 2026-10-29<br>retail price band US$1100-1800 per kg (computed); grade reseller. Bulk 95WNiFe rod US$58-65/kg, 5 kg MOQ is an unverified marketplace listing (estimate) | SPEC |
| S8-TUNGSTEN-PUTTY | Maximum Velocity (reseller); Pinewood Pro US$10.95 (2026). *Tungsten putty 1 oz, minimum density 10 g/cc, US$6.25*. https://maximum-velocity.com/product/tungsten-putty-1-ounce; read 2026-10-29<br>putty density and price for fine-trim ballast | SPEC |
| S8-LEAD | Expert Market Research (Q2 2026); Argus Media (12 Jan 2026); density from standard handbooks (not re-read) (2026). *Lead ingot US$0.89-1.17 per lb; density 11.34 g/cc*. https://www.argusmedia.com/metals-platform/price/assessment/lead-ingot-primary-99-97--del-us-transaction-prices-PA00229370000; read 2026-10-29<br>commodity price band (reseller grade); density grade estimate | SPEC |
| S8-SK78-LINE | Murray's (Robline mini spool, 880 lb); Binks Marine (Liros 410 kg) (2026). *SK78 12-strand 2 mm line: break 399-410 kgf, 1.8 g/m*. https://murrays.com/products/sk78-mini-spool-2mm-dyneema-lashing-line; read 2026-10-29<br>reference strength 3.9 kN at 2 mm, scaled with d^2 for the strength member (scaling estimate) | SPEC |
| S8-POLOLU-4756 | Pololu; Core Electronics listing (2026). *Pololu 131:1 metal gearmotor 37Dx73L mm 12 V, 64 CPR encoder (#4756): 76 rpm, 45 kg-cm stall (extrapolated)*. https://www.pololu.com/product/4756/specs; read 2026-10-29<br>added drive 37d_131 and the 2x variant for heavier sensors (grade reseller for the search summary) | SPEC |
| S8-DRTRADE | team analysis (Claude), 14 Oct 2026 (2026). *DB-01 D&R trade inputs and report (dr_trade.toml, dr_trade_report.txt)*. REPO/analysis/inputs/dr_trade.toml; read 2026-10-29<br>drives, parts masses, tension, drum sizing checks (imported read-only through s5_tow) | COMPUTED |
| S8-BARROWMAN | J. Barrowman, B. Barrowman (1967). *The Theoretical Prediction of the Center of Pressure (Barrowman and Barrowman), NARAM-8*. https://www.apogeerockets.com/education/downloads/Barrowman_Report.pdf; read formulae applied from the method; paper not re-read this session<br>nose CNa 2, CP 0.466 L (ogive); 4-fin trapezoid CNa and CP; subsonic | METHOD |

| SOURCE | REFERENCE AND USE | NATURE |
|---|---|---|
| S8-NACA-TM832 | NACA (1937). *NACA TM 832 towed-body guidance (heavy, compact, damped, small fins far aft), as cited in brief section 5.* not re-read; cited in the orchestrator brief; read not read by S8<br>design guidance only; no number taken from it | METHOD |